INSURANCE COSTS

# Insurance Cost Exploratory Data Analysis

A descriptive review of age, body mass index, smoking status, and annual medical charges

## Summary

The dataset contains 1,338 records and no missing values across its seven columns. Annual charges are strongly right-skewed: the median is $9,382, below the mean of $13,270. Smoking status has the strongest association with charges among the variables examined. Smokers average $32,050 compared with $8,434 for non-smokers; the correlation between charges and a binary smoking indicator is 0.79. The smoking-group difference is the most prominent pattern in the charts and correlations.

## Dataset at a glance

| Measure | Value |
|---|---|
| Records | 1,338 |
| Columns | 7 |
| Missing values | 0 |
| Age | 18–64 years; mean 39.2 |
| BMI | 16.0–53.1; mean 30.7 |
| Annual charges | $1,122–$63,770; median $9,382 |

## Analysis overview

The report reviews the distribution of age, BMI, and charges, compares charges across sex and smoking groups, and examines how charges vary with age and BMI. Histograms summarize how frequently values occur, scatter plots show each record, and the heatmap summarizes pairwise linear relationships. Pearson correlation is used throughout; smoking status is encoded as no = 0 and yes = 1.

# Distributions

Age runs from 18 to 64 years, averaging 39.2. BMI averages 30.7 and ranges from 16.0 to 53.1. Charges have a long upper tail: the middle half range from $4,740 to $16,640.

The age histogram shows a broad spread of policyholders from 18 to 64 years, with records appearing across the full range. The BMI histogram is concentrated around the low 30s, with a longer upper tail. The mean BMI is 30.7 and the median is 30.4, so the center of the distribution is close to 30.

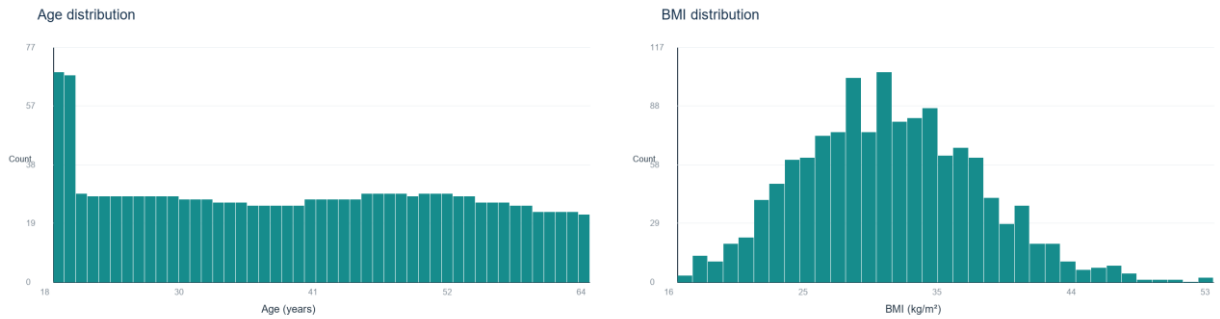



*Figure 1. Age and BMI distributions across all records. Bar height represents the number of records in each interval.*

## Charges by smoking status

The two groups occupy different parts of the charge scale. Smokers (n = 274) have a median charge of $34,456; non-smokers (n = 1,064) have a median of $7,345. The upper histogram panel compares the full distributions, while the lower boxplots make the difference in medians and middle ranges easier to see.

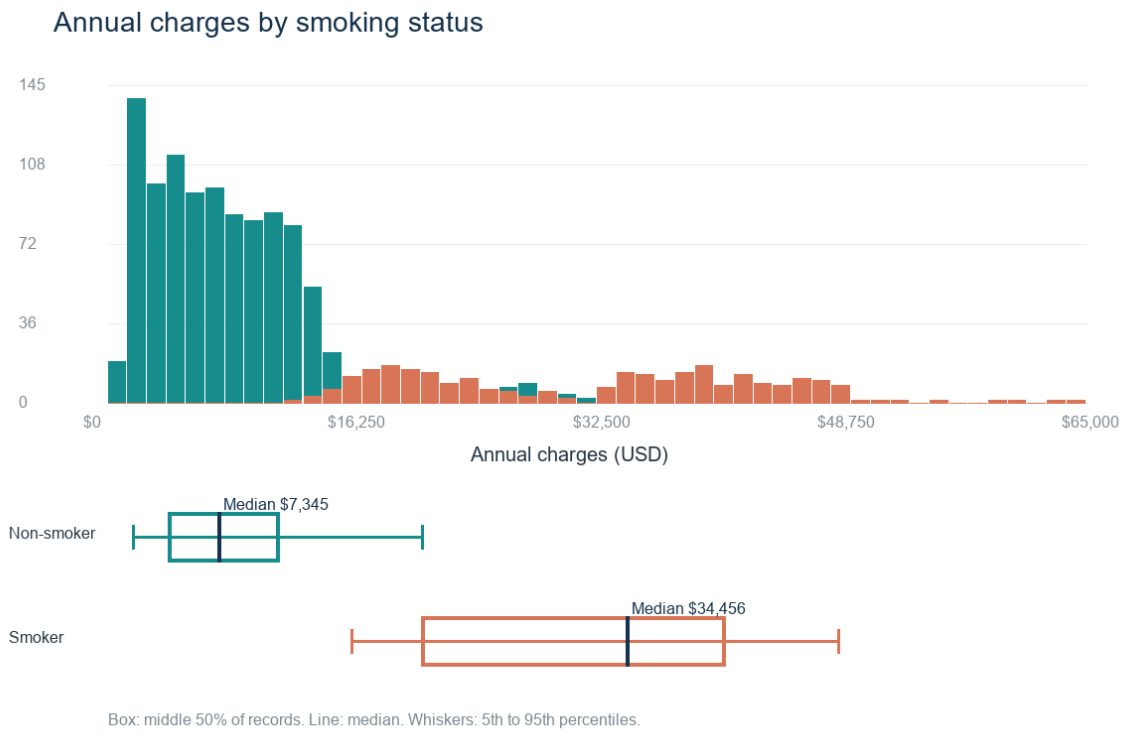



*Figure 2. Annual charges grouped by smoking status. Histograms compare charge frequencies; boxplots show the median and middle 50%, with whiskers from the 5th to 95th percentiles.*

## Group comparisons

Male records have a modestly higher mean charge than female records, while the medians are nearly equal. This gap between means and medians indicates that high-cost observations affect the averages. Regional average charges vary, but these unadjusted comparisons do not account for differences in group composition.

| Sex | Records | Mean charges | Median charges |
|---|---|---|---|
| Female | 662 | $12,570 | $9,413 |
| Male | 676 | $13,957 | $9,370 |

| Region | Records | Mean charges | Median charges |
|---|---|---|---|
| Northeast | 324 | $13,406 | $10,058 |
| Northwest | 325 | $12,418 | $8,966 |
| Southeast | 364 | $14,735 | $9,294 |
| Southwest | 325 | $12,347 | $8,799 |

The sex distributions overlap considerably. Mean charges are $12,570 for female records and $13,957 for male records, while the medians are close ($9,413 and $9,370). In contrast, the smoker comparison shows a much wider shift toward higher charges. Read the two panels by comparing their shapes and locations along the same dollar scale.

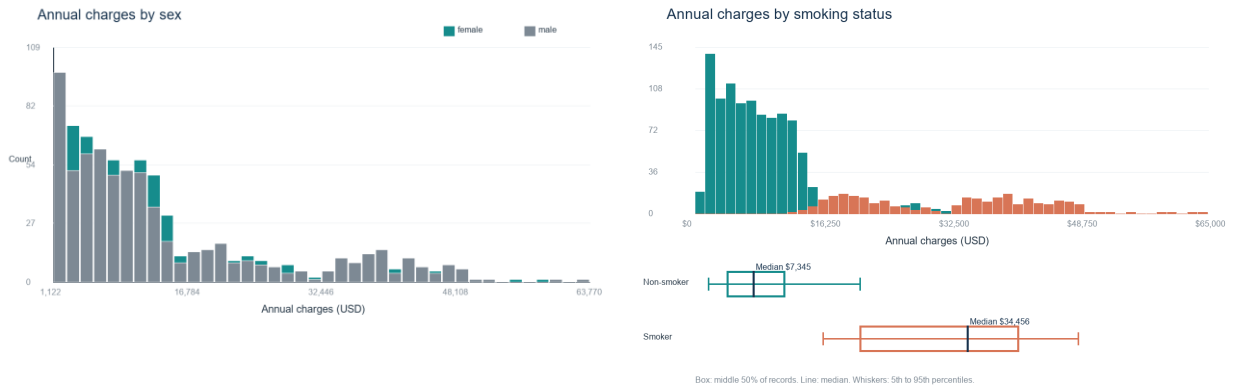



*Figure 3. Charge distributions by sex and smoking status. Each color represents a group; the horizontal axis is annual charges in USD.*

## Age, BMI, and charges

Each point represents one record, with color identifying smoking status. Charges generally rise with age, although values vary widely among people of the same age. The BMI chart shows a weaker upward pattern and especially broad variation at higher BMI values. In both charts, many of the highest charges belong to the smoker group. The panels make it possible to see these patterns at the individual-record level rather than only through group averages.

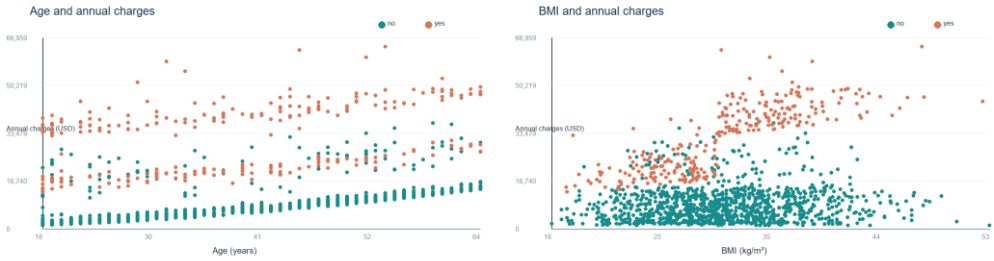



*Figure 4. Annual charges against age and BMI. Each dot is one record; color shows smoking status.*

## Correlation results

The heatmap reports Pearson correlation coefficients from -1 to +1. Positive values indicate that two variables tend to increase together; values near zero indicate little linear association. The diagonal is 1.00 because each variable is perfectly correlated with itself. The smoker column treats yes as 1 and no as 0.

| Variable | Correlation with charges |
|---|---|
| Age | 0.299 |
| BMI | 0.198 |
| Smoker (yes = 1) | 0.787 |
| Children | 0.068 |

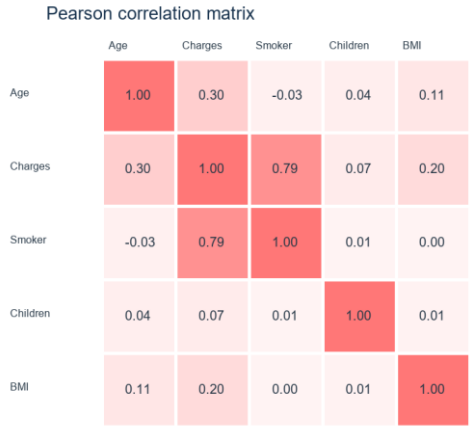



*Figure 5. Pearson correlation matrix. Smoker is encoded as 0/1.*